

Breaking the Vigenère Cipher

Unknown Language

Breaking Vigenère

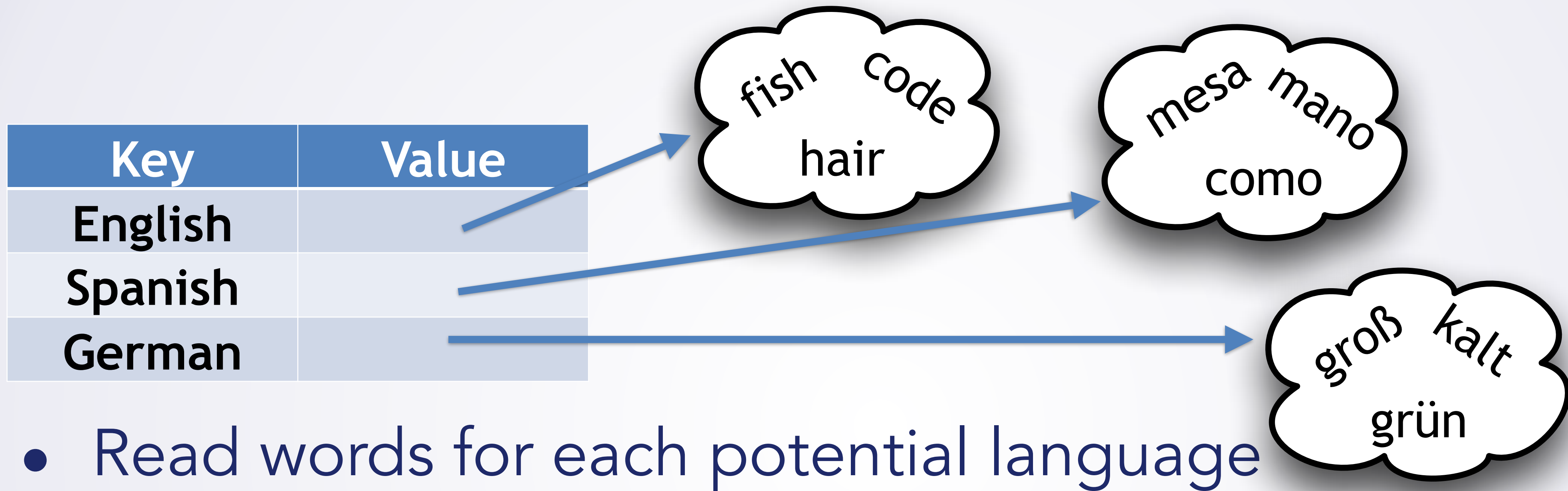
- Step 3: Unknown Language
- For each language, need:
 - Word list
 - Most common character
- Can then try breaking
 - As before: most real words
 - Maximize word count across languages

General Plan

Key	Value
English	
Spanish	
German	

- Read words for each potential language
 - Use `readDictionary` from before
- Make `HashMap<String, HashSet<String>>`
 - Key: language name
 - Value: result of `readDictionary`

General Plan



- Read words for each potential language
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 - Key: language name
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Composition: Complicated Types

HashMap<String, HashSet<String>>

- Remember: Composition
 - Can put pieces together

Plan for Coding

- Two new methods:
 - `char mostCommonCharIn(HashSet<String> words)`
 - `public void breakForAllLangs(String encrypted, HashMap<String, HashSet<String>> languages)`
- Modify two old methods:
 - `public void breakVigenere ()`
 - `public String breakForLanguage(String encrypted, HashSet<String> dictionary)`

Details of What to Do

`char mostCommonCharIn(HashSet<String> words)`

- 'e' not always most frequent
- Count frequency of letters in dictionary
- Proficient in counting occurrences, max

Details of What to Do

```
public void breakForAllLangs(String encrypted,  
    HashMap<String, HashSet<String>> languages)
```

- Try each language in languages.keySet()
 - Use breakForLanguage
 - How many words did it end up with?
 - Pick best language + its decryption

Details of What to Do

public void breakVignere ()

- Method you call from BlueJ
 - Read **all** languages' dictionaries
 - Call **breakForAllLangs** instead of **breakForLanguage**

Details of What to Do

```
public String breakForLanguage(String encrypted,  
HashSet<String> dictionary)
```

- Use mostCommonCharIn instead of 'e'
 - Find right letter for this language